

Notes: Fails because the lower point is more than 2%. We

Customer	BHP
Instrument	GACAL 1100
ID No.	12-0885

Calibration Performed by	Khanh Nguyen
Date	20-11-12
Location	Roxby Down

Audit Equipment

Flow Calibrator Model	TE Number	Cal Due	Pass/Fail
Gacsl 1000	102	15-02-13	PASS

Check Flow Audit equipment STP settings match GasCal settings

Instrument Parameters

System Status OK? (Y/N)	y
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I Lamp Constant	
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MFC 1 Coefficient 0	50.2818
MFC 1 Coefficient 1	2080.4300
MFC 1 Coefficient 2	-3.1256
MFC 2 Coefficient 0	
MFC 2 Coefficient 1	
MFC 2 Coefficient 2	
MFC 3 Coefficient 0	0.0097
MFC 3 Coefficient 1	10.2208
MFC 3 Coefficient 2	-0.0118
MFC 4 Coefficient 0	
MFC 4 Coefficient 1	
MFC 4 Coefficient 2	
O3 Coeff	
O3 Coeff	
O3 Coeff	

D/A Scale (1 ± 0.1)	
O3 Maximum volts	
O2 Minimum volts	
O3 Flow	
Chassis Temperature (°C)	37.62
Lamp Temperature (°C)	
+5 Volts	5.0851
+12 Volts	12.164
-12 Volts	-11.91
+15 Volts	14.886
-15 Volts	-14.91

* Note: MFC 1 = Zero MFC
MFC 2 = Ozone MFC
MFC 3 = Span 1 MFC
MFC 4 = Optional second span flow MFC

Standard Temp (°C) *	0
Standard Pressure (mBar)	1013.25

* Check GasCal standard temp is 0°C

MFC 1 (zero air) Audit

Serial Number	na
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MFC Range (sccm)	1000
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Point	Set Flow (sccm)	Measured Flow (STP)	GasCal Displayed Flow (sccm)	Error (%)
90% FS	9000.00	9018.00	9080.00	-0.68%
80% FS	8000.00	8018.00	8032.00	-0.17%
60% FS	6000.00	6012.00	5993.00	0.32%
40% FS	4000.00	4014.00	3998.00	0.40%
20% FS	2000.00	2012.00	2060.00	-2.33%

Note: Error must be <2% for a pass

MFC Calibration Required

PASS/ FAIL

FAIL

MFC 3 (Span 1) Audit

Serial Number	na	MFC Range (sccm)	50
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Point	Set Flow (sccm)	Measured Flow (STP)	GasCal Displayed Flow	Error (%)
90% FS	50.00	49.83	49.79	0.08%
80% FS	40.00	39.94	39.98	-0.10%
60% FS	30.00	29.94	30.20	-0.86%
40% FS	20.00	19.94	20.27	-1.63%
20% FS	10.00	9.90	10.24	-3.28%

Note: Error must be <2% for a pass

MFC Calibration Required

PASS/ FAIL	FAIL
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MFC 4 (Span 2) Audit

Serial Number		MFC Range (sccm)	
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Point	Set Flow (sccm)	Measured Flow (STP)	GasCal Displayed Flow	Error (%)
90% FS				N/A
80% FS				N/A
60% FS				N/A
40% FS				N/A
20% FS				N/A

Note: Error must be <2% for a pass

PASS/ FAIL	N/A
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MFC 2 (Ozone) Audit

Serial Number		MFC Set-point (sccm)	
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Point	Set Flow (sccm)	Measured Flow (STP)	GasCal Displayed Flow	Error (%)
Set Point				N/A

Note: Error must be <2% for a pass

PASS/ FAIL	N/A
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Calibrator must be returned with this form to head office for Maintenance and Calibration

Technicians Signature	Date
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